KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION

PrepPilot:
AI-Powered Placement Preparation Platform

PROJECT DOCUMENTATION

Project Title: PrepPilot - Placement Prep Coach
Project Type: Full-Stack AI-Powered Web Application (RAG-Based System)
Live Website: https://frontend-ten-blue-59.vercel.app/
GitHub Repository: https://github.com/Harika2003-tech/icl-gen-ai-project-work

Student Name	Registration Number	Role
S Niveditha Krishna	99230040167	RAG Architect
Tanguturi Harika	99230040174	Backend Developer
Pemmaka Hari Priya	99230040153	Frontend Developer
Ummadi Bhavya sree	99230040179	RAG Developer
Boyapati Biindhu Sree	99230040088	LLM Integration

Project Guide / Trainer: Shriwin Prince

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1. Introduction

Placement preparation requires students to practise aptitude, learn technical concepts, improve coding skills and prepare for technical and HR interviews. Using separate platforms for these activities can make preparation difficult to organize and progress harder to monitor.

PrepPilot is a web-based placement preparation platform that brings these activities into one workspace. It includes modules for aptitude practice, Coding Lab with AI-assisted feedback, Technical Prep, HR Prep, AI Mentor, Mock Interview and Progress tracking. The application uses a React frontend and Python FastAPI backend, Firebase for authentication and the Gemini API for AI-assisted features.

2. Problem Statement and Objectives

Problem Statement: Students often switch between different websites and tools to practise aptitude, revise technical topics, improve coding skills and prepare for interviews. This fragmented process can make preparation less organized.

Objectives

• Provide a centralized platform for placement preparation.
• Offer aptitude practice and coding exercises in one workspace.
• Provide AI-assisted feedback and learning support.
• Support technical and HR interview preparation.
• Help learners review preparation activity through progress tracking.
• Deploy an accessible web application with authenticated user access.

3. System Architecture and Workflow

PrepPilot follows a frontend-backend architecture. The frontend presents the user interface and sends API requests to the backend. The backend handles application endpoints and coordinates AI-assisted operations. Firebase provides authentication, while the Gemini API supports configured generative AI features.

3.1 Main Application Flow

1. User access: The user opens the web application and signs in through the authentication interface.
2. Module selection: The user selects Aptitude, Coding Lab, Technical Prep, HR Prep, AI Mentor, Mock Interview or Progress.
3. API request: The frontend sends requests to the FastAPI backend using the configured API base URL.
4. Backend processing: The backend processes the request and, where applicable, calls the configured Gemini model.
5. Response display: The backend response is returned to the frontend and displayed to the user.

3.2 AI-Assisted Features

The AI Mentor provides conversational learning support. The Coding Lab can send submitted code for AI-assisted review and feedback. It should be described as code review rather than online code execution unless a code execution engine is separately implemented and verified.

3.3 Deployment Architecture

The frontend is deployed on Vercel and the backend is deployed on Render. The frontend uses the VITE_API_URL environment variable to communicate with the backend. FastAPI CORS middleware allows requests from the deployed frontend origin.

4. Implementation Details and Technology Stack

Component	Technology	Purpose
Frontend	React, Vite, JavaScript	User interface and client-side application
UI / Styling	CSS / Tailwind CSS (project configuration)	Layout and visual presentation
Backend	Python, FastAPI, Uvicorn	API endpoints and backend processing
Authentication	Firebase Authentication	User sign-in and authentication
Generative AI	Google Gemini API (configured model: gemini-3.1-flash-lite)	AI Mentor and AI-assisted responses
Frontend hosting	Vercel	Hosting the web frontend
Backend hosting	Render	Hosting the FastAPI service

4.1 Major Modules

• **Overview:** Main workspace and preparation summary.
• **Aptitude:** Placement aptitude practice questions.
• **Coding Lab:** Coding practice with AI-assisted code feedback.
• **Technical Prep:** Technical learning and revision support.
• **HR Prep:** HR interview preparation.
• **AI Mentor:** AI-based conversational assistance.
• **Mock Interview:** Interview practice.
• **Progress:** Preparation progress and activity review.

5. Results and Testing

The application frontend is hosted on Vercel and the backend is hosted on Render. During deployment verification, the backend health endpoint returned a successful status and indicated that Gemini and Firebase configuration were present. A CORS issue initially prevented browser requests from the Vercel frontend from reaching the backend. The backend CORS allowlist was updated, the change was committed and pushed to GitHub, and Render successfully deployed the updated commit.

After the CORS correction, the user confirmed that the application modules were displaying and working. The production URL retained for use is:

Production URL: https://frontend-ten-blue-59.vercel.app

Backend URL: https://preppilot-backend-df7x.onrender.com

5.1 Testing Checklist

Test Area	Verification	Status
Backend health	Health endpoint returned status ok	Passed
Frontend-backend connection	CORS allowlist updated and dashboard requests retested	Passed after correction
Aptitude	Questions displayed after deployment fix	User-confirmed working
Other modules	User reported modules were displaying and working	User-confirmed working

6. Conclusion and Future Scope

Conclusion: PrepPilot brings multiple placement preparation activities into a single web application. It combines a React frontend, FastAPI backend, Firebase authentication and Gemini-powered AI assistance. The application has been deployed using Vercel and Render. The CORS issue encountered during deployment was corrected and the user verified that the modules were working.

Future Scope: Possible improvements include expanding the aptitude question bank, adding more coding problems and structured test cases, improving personalized progress analytics, adding topic-wise learning paths, strengthening interview evaluation and feedback, and improving persistence of user progress and uploaded learning resources.